4.)

$$x - 2y + 3z = 9$$
$$2x + 4y = 9$$
$$3x - 2z + 3 = 0$$

a. Harga x metode Cramer

$$D_x = \begin{vmatrix} 9 & -2 & 3 \\ 9 & 4 & 0 \\ -3 & 0 & -2 \end{vmatrix} \begin{matrix} 9 & -2 \\ 9 & 4 \\ -3 & 0 \end{matrix} = -72 + 0 + 0 - (-36 + 0 + 36)$$
$$= -72$$

$$D_y = \begin{vmatrix} 1 & 9 & 3 \\ 2 & 9 & 0 \\ 3 & -3 & -2 \end{vmatrix} \begin{matrix} 1 & 9 \\ 2 & 9 \\ 3 & -3 \end{matrix} = -18 + 0 - 18 - (81 + 0 - 36)$$
$$= -36 - 45 = -81$$

$$D_z = \begin{vmatrix} 1 & -2 & 9 \\ 2 & 4 & 9 \\ 3 & 0 & 3 \end{vmatrix} \begin{matrix} 1 & -2 \\ 2 & 4 \\ 3 & 0 \end{matrix} = -12 - 54 + 0 - (108 + 0 + 12)$$
$$= -66 - 120 = -186$$

Dipindai dengan CamScanner

$$D = \begin{vmatrix} 1 & -2 & 3 \\ 2 & 4 & 0 \\ 3 & 0 & -2 \end{vmatrix} \begin{matrix} 1 & -2 \\ 2 & 4 \\ 3 & 0 \end{matrix} = -8 + 0 + 0 - (36 + 0 + 8)$$

$$= -8 - 44 = -52$$

Jadi, $x = \frac{Dx}{D} = \frac{-72}{-52} = \frac{18}{13}$

$$y = \frac{Dy}{D} = \frac{-84}{-52} = \frac{81}{52}$$

$$z = \frac{Dz}{D} = \frac{-186}{-52} = \frac{93}{26}$$

6. Metode Invers matriks

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 & -2 & 3 \\ 2 & 4 & 0 \\ 3 & 0 & -2 \end{pmatrix}^{-1} \cdot \begin{pmatrix} 9 \\ 9 \\ -3 \end{pmatrix}$$

$$= \begin{pmatrix} 2/13 & 1/13 & 3/13 \\ -1/13 & 11/52 & -3/26 \\ 3/13 & 3/26 & -2/13 \end{pmatrix} \begin{pmatrix} 9 \\ 9 \\ -3 \end{pmatrix}$$

$$= \begin{pmatrix} 18/13 \\ 81/52 \\ 93/26 \end{pmatrix}$$

Dipindai dengan CamScanner